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Car seat base with installation guidance and locking anti-rebounding mechanism

Abstract

A system for guiding installation of car seat base and a corresponding car seat includes sensors mounted on the car seat base for detecting a progress in a plurality of sequential installation steps. The installation steps are required for installing the car seat base and the corresponding car seat on a vehicle seat. The car seat base further includes light emitting elements arranged at various locations on the car seat base, each location corresponding to one of the installation steps. A processor is adapted to execute code instructions for, in response to outputs of the sensors, identifying completion of a current installation step. The processor is further adapted to instruct a suitable light emitting element to indicate a location corresponding to an installation step following the current installation step. An anti-rebound device includes a spring or piston that adaptively adjusts in response to an impact and a locking mechanism.

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Background/Summary

RELATED APPLICATIONS (1) This Application is a National Phase of PCT Patent Application No. PCT/IL2020/051142 having International filing date of Nov. 4, 2020, which claims the benefit of priority under 35 USC § 119 (e) of U.S. Provisional Patent Application No. 62/930,012, filed on Nov. 4, 2019. The contents of the above applications are all incorporated by reference as if fully set forth herein in their entirety.

FIELD AND BACKGROUND OF THE INVENTION
(1) The present invention, in some embodiments thereof, relates to a base for a car seat for a vehicle, and more particularly, but not exclusively, to mechanisms for improving ease of installation of a car seat base. The present invention also relates to an anti-rebounding mechanism for a car seat base with a locking feature, and to a mechanism for improving ease of installation of such an anti-rebounding mechanism.
(2) Car seats have a high occurrence of misuse due to installation complexity. This is because car seats are highly mechanical products that need to be connected to a car in various positions, comply with various standards, adjust to different car seat designs and be installed in accordance with a child's size. The positions may include: rear-facing attached with seat belt; rear facing attached with Isofix latch; front-facing attached with seat belt; front facing attached with Isofix latch; a combination thereof (seat belt and Isofix), installation with or without a separate base; installation with or without a top tether; and installation with or without a load-bearing leg. Due to the complexity and number of installation steps, many of the steps are not done correctly, putting a child at risk.
(3) Existing car seats partially address these concerns by providing indications that certain installation steps have been performed correctly. For example, upon successful completion of an installation step for a car seat, the car seat may display a light signal. Alternatively, the car seat base may include an indicator that is partially colored red and partially colored green, and which is mechanically movable relative to a window. The red portion of the indicator may be visible through the window prior to completion of the installation step. Completion of the installation step may move the indicator such that the green portion is now visible through the window. Other indications include emitting an audible sound, such as a “click,” when the step is completed.
(4) Car seat bases may be attachable to a vehicle with one or more of two complementary processes. In one process, the car seat base may include a seat belt guide. The seat belt of the vehicle is guided through the guide and buckled, thus retaining the base against the vehicle seat. In a second process, connectors attached to the car seat base are attached to fittings built into the vehicle seat. This process is known as Isofix or LATCH, and has been standardized as International Safety Organization standard ISO 13216.
(5) Car seats may also feature anti-rebounding devices. An anti-rebounding device is an element laying against the seat back cushion (near the baby's feet, in a rear-facing car seat) that minimizes the rebound during and after a crash. Rebound is movement after the initial crash. For rear-facing babies, since the head area of their car seat is typically not braced against anything, the rebound usually causes the car seat bounce up and back toward the vehicle seatback. The rebound forces

can be severe enough to allow a rear-facing child to strike the vehicle seatback. An anti-rebound device may be a bar installed at the foot of the car seat or at the rear of the base that braces the car seat against the seat back.

(6) An exemplary prior art car seat base is illustrated in FIG. **1**. Car seat base **1** includes anti-rebounding device **2**. The anti-rebounding device **2** is capable of being raised only to a fixed position. In order to allow the car seat base **1** to be installed in vehicle seats of different sizes, the anti-rebounding device **2** is laterally adjustable along arrow **3**. Car seat base **1** also includes Isofix latches **4**. In order to allow the car seat base **1** to be installed on vehicle seats with different sizes, the Isofix latches **4** are also laterally adjustable along arrow **5**.

SUMMARY OF THE INVENTION

(7) The above-described indications for installation steps provide guidance to the user only after an installation step has been completed. They do not provide a user any prospective guidance as to which steps to complete. Accordingly, a user must consult other sources of instruction, such as an installation manual or a sticker mounted on the car seat base. The use of an installation manual or sticker may cause unnecessary confusion and may promote installation errors in the event that the manual or sticker is not available (e.g., they were lost or defaced).

(8) In addition, installation steps for car seat bases may proceed from an initial step of either buckling the base to the vehicle seat with a seat belt or attaching Isofix connectors to fittings installed on the vehicle. It would be desirable for subsequent installation steps to proceed intuitively and identically, whether proceeding from a buckling step or from an Isofix connection step.

(9) In addition, it is desirable to provide a car seat base that is safe and intuitive to install, including an anti-rebound device that maintains its position regardless of the severity of an impact.

(10) The disclosed embodiments provide a car seat base that exhibits numerous advantages, including those described above.

(11) The disclosed embodiments of a car seat base provide automated guidance to a user regarding progress in installation of the car seat. This guidance takes the form of light indications in specific areas of the car seat base, a pattern of light indications at a centralized portion of the car seat base, and/or an indication on a mobile application on the user's mobile device, or on a vehicle screen, or any other mobile or screen option. Prior to completion of each installation step, a light indication is displayed in the specific area of the car seat base that needs to be installed, and visual feedback and real-time instructions are provided in the application. After the installation step is performed correctly, an assurance indication is shown to the user, both by a light signal on the base and through a message or other indication in the application.

(12) In addition, the car seat base is installed in just four steps. These four steps are performed intuitively and with a minimum of adjustments by the user. In a first step, a user affixes the base to the vehicle, either through an Isofix system or a vehicle seat belt. Unlike the prior art car seat base depicted in FIG. **1**, the car seat base is equipped with rigid Isofix latches, so that the user does not need to open the latches and adjust their lateral locations. The car seat base is equipped with a handle. When the user carries the base from the handle (for example, to its position on the vehicle seat), a load leg drops down from a folded position underneath the car seat base to an extended position. The load leg is equipped with an angle sensor that informs the user the angle of the leg in real time, and a telescopic extension for anchoring the load leg to the floor of the vehicle. Thus, when the user attaches the base to the vehicle, the leg is already dropped down.

(13) In a second step, the user extends an anti-rebounding device. The anti-rebounding device is self-extracting and self-adjusting to the back of the vehicle seat. Unlike the prior art car seat base depicted in FIG. **1**, the anti-rebound device need not be adjusted manually to meet the back of the vehicle seat.

(14) In a third step, the user extends the telescopic extension of the load leg to the floor of the vehicle. In a fourth step, the user attaches the car seat to the car seat base.

(15) It is not possible to attach the car seat to the car seat base unless the car seat base is properly connected to the vehicle and the rebounder is in position. Furthermore, aside from controlling the extension of the leg, the user has no discretion in locating components of the car seat base during the installation process. As a result, the car seat base is installed intuitively and safely.
(16) According to a first aspect, a system for guiding installation of a car seat base and a corresponding car seat includes sensors mounted on the car seat base for detecting a progress of a user in a plurality of sequential installation steps. The installation steps are required for installing the car seat base and the corresponding car seat on a vehicle seat. The car seat base further includes light emitting elements arranged at various locations on the car seat base, each location corresponding to one of the plurality of installation steps. A processor is adapted to execute code instructions for, in response to outputs of at least one of the plurality of sensors, identifying completion of a current installation step of the plurality of installation steps. The processor is further adapted to execute code instructions for instructing a suitable light emitting element from the plurality of light emitting elements to indicate a location corresponding to an installation step following the current installation step.
(17) Advantageously, the car seat base directs a user to perform each sequential installation step following completion of prior installation steps. A user need only pay attention to the light signals, in order to both verify that a prior installation step is successfully completed, and to determine which section of the car seat base to install next.
(18) In another implementation according to the first aspect, a central light interface is comprised of light emitting elements arranged in a pattern. Each of the light emitting elements of the central light interface corresponding to one of the installation steps. For each of the installation steps, whenever a light emitting element indicates a location corresponding to an installation step, a corresponding light from the central light interface displays a first type of light signal. Whenever a sensor senses that said particular installation step is complete, the corresponding light displays a second light signal. Advantageously, the central light interface provides a user with visual indication when each installation step is complete, as well as indication of progress in the entire installation process, through display of the light emitting elements in the pattern. Thus, at each stage of the installation, a simple glance is sufficient to understand the installation status and its accuracy.
(19) In another implementation according to the first aspect, upon detection of an error status by one of the sensors, one or more of the light emitting elements is configured to display an error light signal to instruct the user to rectify the error. Optionally, the error includes at least one of: (1) installation step not completed correctly or completed in the wrong sequence; (2) child not buckled; (3) one of the parts of the car seat base or car seat not properly connected; and (4) car seat no longer structurally safe due to accident impact. Advantageously, the car seat base thus detects errors and alerts the errors to users.
(20) In another implementation according to the first aspect, a communication module is configured to wirelessly deliver information to a mobile device about a status of each of the installation steps and/or an alert status. Code instructions are embodied in a non-transitory medium stored on the mobile device. The code instructions are configured, upon receipt of a status of completion of each installation step, to display a pictorial representation of a progress of installation of the car seat base and instructions for performing a subsequent step of the sequential installation steps. Advantageously, the car seat base thus delivers guidance to the user even when the user is not physically observing the car seat base. In addition, the guidance provided through the mobile device may be more detailed than the guidance provided by the lights on the car seat base.
(21) In another implementation according to the first aspect, the car seat base includes a speaker for delivering audible messages regarding a status of each of the installation steps and/or an alert status. Advantageously, the speaker provides an audible message that complements the visual

messaging provided by the light emitting elements.

(22) In another implementation according to the first aspect, the sensors include sensors for detecting one or more of: (1) leg angle of a telescopic load leg; (2) pressure on the telescopic load leg; (3) extension of the telescopic load leg; (4) whether an Isofix connector of the base is engaged with a corresponding child safety seat attachment point of the vehicle; (5) whether a seat belt is passed through a seat belt guide; (6) whether an anti-rebound device is in a folded position or an extended position; (7) orientation of car seat as front facing or rear facing; (8) whether a car seat belt buckle is engaged; (9) presence of a child in the car seat; (10) reclining angle of the car seat; (11) type of car seat connected to the base; (12) whether the child car seat was subjected to substantial forces and is not suitable for usage or installation; and (13) whether a power source for the car seat base has sufficient power. Advantageously, the car seat base may thus provide information and instruction regarding the outputs of each of these sensors.

(23) In another implementation according to the first aspect, the car seat base further includes a frame, a load leg that is pivotable from a folded position beneath the frame to an extended position between the frame and a vehicle floor, and at least one handle or grip oriented on the frame to enable grasping of the frame. When the car seat base is grasped from the at least one handle or grip, the leg pivots into the extended position. The presence of the handle, grip area, or combination thereof, thus assists a user in placing the car seat base onto a seat of the vehicle while simultaneously ensuring that the leg is correctly placed.

(24) In another implementation according to the first aspect, the at least one handle or grip is configured on at least one of (1) on an upper portion of the frame; (2) on a front portion of the frame; (3) in a rear portion of the frame; and (4) on two parallel side portions of the frame.

(25) In another implementation according to the first aspect, a locking mechanism locks the leg into the extended position. Advantageously, the locking mechanism maintains the leg in the extended position and prevents accidental refolding of the leg into the folded position.

(26) According to a second aspect, a car seat base includes a frame shaped to be mounted on a vehicle seat, and having distal and proximal ends. The car seat base has a release mechanism. A rod is mechanically connected to (1) a stopper preventing release of the release mechanism; (2) a belt pedal adapted to be located beneath a seat belt; and (3) two arms each having a connecting latch. A depression of the belt pedal by the seat belt when the seat belt is bucked to the vehicle, or a locking of the connectors to child safety seat attachment points of the vehicle seat, induces a mechanical retraction of the stopper, thereby allowing release of the release mechanism. Advantageously, the release mechanism thus is released in the same way, whether the car seat base is initially attached to the vehicle with a seatbelt or with an Isofix mechanism.

(27) In another implementation according to the second aspect, the release mechanism includes a push button release mechanism, the stopper is a button stopper preventing depression of the push button, and the depression of the belt pedal or the locking of the connecting latches induces a mechanical retraction of the button stopper, thereby allowing depression of the push button.

(28) In another implementation according to the second aspect, an anti-rebound device is fixedly connected to the distal end and releasably connected to the proximal end. The anti-rebound device is convertible between (1) a folded position wherein the anti-rebound device is parallel to the frame, and (2) an extended position wherein the anti-rebound device forms a wide angle with the frame and contacts a back rest of the vehicle seat, to thereby limit rebounding of the car seat base. The release mechanism releases the anti-rebound device from the folded position into the extended position. Optionally, the anti-rebound device is released automatically following the depression of the belt pedal or the locking of the connecting latches. Advantageously, the anti-rebound device is thus installed in an intuitive manner. In addition, it is possible to convert the anti-rebound device from the folded position to the extended position only after the car seat base has been properly attached to the vehicle seat, whether via the seat belt or via the Isofix connectors.

(29) In another implementation according to the second aspect, the anti-rebound device is spring-

loaded or piston-loaded and extends from the folded position into the extended position by expansion of the spring or extension of the piston. Advantageously, the spring or piston provide a compact, efficient mechanism for extension of the anti-rebound device.

(30) In another implementation according to the second aspect, the anti-rebound device further comprises a locking mechanism configured to prevent a return of the anti-rebound device from the extended position to the folded position. The locking mechanism includes an upper lock arm having a tapered receptacle at a lower end of said upper lock arm. The tapered receptacle includes a wide end at a lower extent thereof, a narrow end at a higher extent thereof, and at least one angled surface. The tapered receptacle receives therein a lower lock arm configured to slide through the tapered receptacle and a roller held by a spring. When the anti-rebound device is in the extended position, retraction of the lower lock arm causes rotation of the roller toward the narrow end, whereby the spring forces the roller to contact the lower lock arm and at least one of the at least one angled surface, thereby locking the lower lock arm against the roller and the tapered receptacle. Advantageously, this locking mechanism is unidirectional, permitting opening of the anti-rebound device but not allowing closing of the anti-rebound device.

(31) Optionally, the locking mechanism further includes a release pedal mechanically connected to an arm extending to the narrow end of the tapered receptacle. Depression of the release pedal causes the arm to displace the roller from the narrow end, thereby permitting retraction of the lower lock arm through the tapered receptacle. The arm may be oriented substantially parallel to the upper lock arm. Advantageously, the release pedal provides a compact and intuitive system for unlocking the locking mechanism and permitting return of the anti-rebound device to the folded position.

(32) In another implementation according to the second aspect, each of the arms comprises a spring-loaded piston. A first pivot is configured between each of the arms and the rod. Attachment of the connecting latches to the child safety seat attachment points drives the spring-loaded piston forward, thereby pivoting the first pivot in a forward direction and causing corresponding rearward movement of the rod.

(33) In another implementation according to the second aspect, the belt pedal is attached to the rod with a second pivot. An extension is underneath the belt pedal. Depression of the belt pedal by the seat belt drives the extension into the second pivot, thereby pivoting the second pivot in a forward direction and causing corresponding rearward movement of the rod.

(34) In another implementation according to the second aspect, substantially identical rearward movement of the rod is caused by either buckling the seat belt or by attaching the connectors to the fittings. Advantageously, the push button release mechanism is thus released in an identical way, regardless of the method employed to attach the car seat base to a vehicle seat.

(35) According to a third aspect, a car seat base includes an anti-rebound device. A spring or piston automatically extends the anti-rebound device and secures the anti-rebound device against a vehicle seat back rest. The spring or piston are configured to adaptively adjust their configuration so as to secure the car seat base from rebounding in response to an impact. Advantageously, the adaptive adjustment of the anti-rebound device ensures that the anti-rebound device remains in place, even if the base moves or shifts, for example, during an accident. In addition, the adaptive adjustment of the anti-rebound device fixes the car seat in place during an accident, thereby improving the safety of the car seat.

(36) In another implementation according to the third aspect, the car seat base includes a proximal end and a distal end. The anti-rebound device is fixedly connected to the distal end and releasably connected to the proximal end. The anti-rebound device is convertible between a folded position wherein the anti-rebound device is parallel to the frame, and an extended position wherein the anti-rebound device forms a wide angle with the frame and contacts the back rest of the vehicle seat. A push button release mechanism, when depressed, releases the anti-rebound device from the folded position into the extended position. Advantageously, the anti-rebound device may be intuitively and

easily installed by depression of the push button.
(37) In another implementation according to the third aspect, the anti-rebound device further comprises a locking mechanism configured to prevent a return of the anti-rebound device from the extended position to the folded position. The locking mechanism includes an upper lock arm having a tapered receptacle at a lower end of said upper lock arm. The tapered receptacle includes a wide end at a lower extent thereof, a narrow end at a higher extent thereof, and at least one angled surface. The tapered receptacle receives therein a lower lock arm configured to slide through the tapered receptacle and a roller held by a spring. When the anti-rebound device is in the extended position, retraction of the lower lock arm causes rotation of the roller toward the narrow end, whereby the spring forces the roller to contact the lower lock arm and at least one of the at least one angled surface, thereby locking the lower lock arm against the roller and the tapered receptacle. Advantageously, this locking mechanism is unidirectional, permitting opening of the anti-rebound device but not allowing closing of the anti-rebound device.
(38) Optionally, the locking mechanism further includes a release pedal mechanically connected to an arm extending to the narrow end of the tapered receptacle. Depression of the release pedal causes the arm to displace the roller from the narrow end, thereby permitting retraction of the lower lock arm through the tapered receptacle. The arm may be oriented substantially parallel to the upper lock arm. Advantageously, the release pedal provides a compact and intuitive system for unlocking the locking mechanism and permitting return of the anti-rebound device to the folded position.
(39) Other systems, methods, features, and advantages of the present disclosure will be or become apparent to one with skill in the art upon examination of the following drawings and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present disclosure, and be protected by the accompanying claims.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) Some embodiments of the invention are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of embodiments of the invention. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the invention may be practiced.
(2) In the drawings:
(3) FIG. **1** is a side view of a prior art car seat base;
(4) FIG. **2** is a left perspective view of a car seat base with an anti-rebound device in a folded position, according to embodiments of the invention;
(5) FIG. **3** is a top view of the car seat base of FIG. **2**, according to embodiments of the invention;
(6) FIG. **4** is a left perspective view of the car seat base of FIG. **2** with the anti-rebound device in an extended position, according to embodiments of the invention;
(7) FIG. **5**A is a depiction of a central light interface of the car seat base of FIG. **2** during or following attachment of the car seat base to a vehicle seat, according to embodiments of the invention;
(8) FIG. **5**B is a depiction of the central light interface of FIG. **5**A during or following extension of the anti-rebound device, according to embodiments of the invention;
(9) FIG. **5**C is a depiction of the central light interface of FIG. **5**A during or following telescopic extension of a load-bearing leg to a floor of the vehicle, according to embodiments of the invention;

(10) FIG. **5**D is a depiction of the central light interface of FIG. **5**A during or following attachment of a car seat to the car seat base, according to embodiments of the invention;
(11) FIG. **5**E is a depiction of the central light interface of FIG. **5**A during or following buckling a child in the car seat, according to embodiments of the invention;
(12) FIG. **6**A is a depiction of a second embodiment of a central light interface during or following attachment of the car seat base to a vehicle seat, according to embodiments of the invention;
(13) FIG. **6**B is a depiction of the central light interface of FIG. **6**A during or following extension of the anti-rebound device, according to embodiments of the invention;
(14) FIG. **6**C is a depiction of the central light interface of FIG. **6**A during or following telescopic extension of a load-bearing leg to a floor of the vehicle, according to embodiments of the invention;
(15) FIG. **6**D is a depiction of the central light interface of FIG. **6**A during or following attachment of a car seat to the car seat base, according to embodiments of the invention;
(16) FIG. **6**E is a depiction of the central light interface of FIG. **6**A during or following buckling a child in the car seat, according to embodiments of the invention;
(17) FIG. **7**A is a depiction of a third embodiment of a central light interface;
(18) FIG. **7**B is a depiction of a fourth embodiment of a central light interface;
(19) FIG. **7**C is a depiction of a fifth embodiment of a central light interface;
(20) FIG. **8**A is a depiction of an anti-rebound device of the car seat base of FIG. **2**, in a partially extended position;
(21) FIG. **8**B is a depiction of an anti-rebound device of the car seat base of FIG. **2**, in a fully extended position;
(22) FIG. **9**A is a depiction of another embodiment of a car seat base having anti-rebound device with a sleeve covering an extending mechanism and a locking mechanism;
(23) FIG. **9**B is a perspective view of the anti-rebound device of the car seat base of FIG. **9**A with the sleeve;
(24) FIG. **9**C is a perspective view of the anti-rebound device of the car seat base of FIG. **9**A, showing the extending mechanism and locking mechanism with the sleeve removed;
(25) FIG. **10**A is a perspective view of the locking mechanism of the anti-rebound device of FIG. **9**A;
(26) FIG. **10**B is a side view of the locking mechanism of the anti-rebound device of FIG. **9**A;
(27) FIG. **11**A is a close-up depiction of the locking mechanism during extension of the anti-rebound device;
(28) FIG. **11**B is a close-up depiction of the locking mechanism during attempted retraction of the anti-rebound device;
(29) FIG. **12**A is a side view of the locking mechanism, showing a release pedal in a depressed position;
(30) FIG. **12**B is a close up depiction of elements of the locking mechanism with the release pedal in the depressed position;
(31) FIG. **13**A is a schematic depiction of a push button release mechanism of the car seat base of FIG. **2** in a locked position, with a button stopper preventing depression of the push button, according to embodiments of the invention;
(32) FIG. **13**B is a schematic depiction of the push button release mechanism of FIG. **13**A with the button stopper mechanically retracted, allowing depression of the push button, according to embodiments of the invention;
(33) FIGS. **14**A-**14**C are further schematic views of the push button release mechanism in the locked position; and
(34) FIGS. **15**A-**15**C are further schematic views of the push button release mechanism, depicting movement of the push button release mechanism into an unlocked position in which the button stopper does not prevent depression of the push button, according to embodiments of the invention.

DESCRIPTION OF SPECIFIC EMBODIMENTS OF THE INVENTION

(35) The present invention, in some embodiments thereof, relates to a base for a car seat for a vehicle, and more particularly, but not exclusively, to mechanisms for improving ease of installation of a car seat base. The present invention also relates to an anti-rebounding mechanism for a car seat base and to a mechanism for improving ease of installation of such an anti-rebounding mechanism.

(36) Before explaining at least one embodiment of the invention in detail, it is to be understood that the invention is not necessarily limited in its application to the details of construction and the arrangement of the components and/or methods set forth in the following description and/or illustrated in the drawings and/or the Examples. The invention is capable of other embodiments or of being practiced or carried out in various ways.

(37) Referring to FIGS. **2-4**, car seat base **10** is configured to be attached to a vehicle seat. Car seat base **10** is part of a two-part transportation system, along with a corresponding child car seat that holds the child. The reason for having a separate car seat base **10** is that, in some cases, there is a need to take the car seat itself in and out of the vehicle—for example, transferring an infant car seat from the vehicle to a stroller, and vice versa. The car seat base **10** remains permanently installed in the vehicle. Replacing the car seat thus requires only attaching the car seat to the car seat base **10**, making reinstalling the car seat faster, simpler, and safer.

(38) Car seat base **10** is composed of a frame, including lower outer portion **12** and upper outer portion **14**. The frame may include various internal components and safety features, such as those described in the copending international application, filed on herewith date, which claims priority to U.S. Provisional Patent Application 62/930,008, filed Nov. 4, 2019, entitled “Safety Seat Protecting From Side Impact” the copending international application, filed on herewith date, which claims priority to U.S. Provisional Application 62/930,003, filed Nov. 4, 2019, entitled “Methods and Systems for Safety Seat”, the contents of which are incorporated herein by reference as if fully set forth herein. Lower outer portion **12** and upper portion **14** define a distal portion, which is the portion closer to the vehicle seat back when the car seat base **10** is installed, and a proximal portion, which is the portion closer to the front of the vehicle seat when the car seat base is installed.

(39) Lower outer portion **12** includes abutments or arms **16** for Isofix connectors **34**. Isofix connectors **34** are also referred to herein as “connecting latches.” The Isofix connectors **34** are configured to latch onto corresponding fittings that are attached to the vehicle seat. The fittings may also be referred to herein as child safety seat attachment points. In a preferred embodiment, the Isofix connectors **34** are laterally fixed, meaning that the arms **16** are not adjustable in length. An advantage of this embodiment is that the user is unable to make any installation errors when adjusting the length of the arms **16**. In addition, the car seat base **10** is properly secured against the back of the vehicle seat due to the self-extending action of the anti-rebound device, as will be discussed further herein.

(40) The frame also includes seat belt guide **36**. The seat belt guide **36** is in the distal portion of the car seat base, and provides a passageway for threading a seat belt therethrough, for attachment to a buckle.

(41) The upper portion **14** includes an anti-rebound device **18**. Anti-rebound device **18** is fixedly connected to the distal end of upper portion **14**, and is releasably connected to the proximal end. The anti-rebound device **18** is convertible between a folded position (FIGS. **2** and **3**), in which the anti-rebound device **18** is somewhat parallel to the frame **14**, and an extended position (FIG. **4**), in which the anti-rebound device **18** forms a wide angle with the frame and contacts a back rest of the vehicle seat, to thereby limit rebounding of the car seat base **10**.

(42) Push button release mechanism **20** is located on the upper portion **14**, proximal to the anti-rebound device **18**. Depression of the push button **20** releases the anti-rebound device **18** from the folded position into the extended position. In alternative embodiments, the release mechanism for

the anti-rebound device **18** does not include a push button, and the anti-rebound device **18** is automatically released into the extended position upon attachment of the car seat base **10** to the vehicle seat.

(43) Telescopic load leg **22** extends below the proximal portion of frame **12**. Telescopic load leg **22** includes base section **21** and extension section **23**. The extension section **23** is adjustably extendible from the end of base section **21** to the floor of the vehicle.

(44) Telescopic load leg **22** is pivotably attached to the frame, e.g., with a hinge. The load leg **22** is pivotable from a folded position beneath or within the frame **12** to an extended position between the frame **12** and the vehicle floor. Advantageously, when the car seat base **10** is not installed in a vehicle, the telescopic load leg **22** folds into a cavity within frame **12** for compact storage. Load leg **22** may also include hinge lock **29**, which may lock the load leg **22** in the extended position when the car seat base is installed on the vehicle. Hinge lock **29** may be any suitable lock, e.g., a mechanical lock or a magnetic lock.

(45) Car seat base **10** may include one or more handles or grips to assist a user in transporting and placing the car seat base. The handles or grips may be on all faces of the car seat base **10**, including the front, rear, and sides. In the depicted embodiment, handle **24** is formed within the anti-rebound device **18**. When the anti-rebound device **18** is in the folded position, the handle **24** is positioned such that a user may carry the car seat base **10** therewith. In some embodiments, when the user carries the car seat base **10** holding the handle **24**, the telescopic load leg **22** (at this point, with leg extension **23** not extended) automatically pivots into the extended position. Once pivoted into the extended position, the load leg **22** may be locked by hinge lock **29** so as to stay in the extended position. Advantageously, in such embodiments, a user bringing the car seat base **10** to the vehicle while grasping the handle **24** will naturally place the car seat base **10** onto the vehicle seat with the telescopic leg **22** in its unfolded position. The handle **24** thus, in addition to providing an aid in transport, also helps prevent an installation error, e.g., in which the car seat base **10** is attached to the vehicle seat while the telescopic leg **22** is folded underneath or within the frame **12**.

(46) Car seat attachment points **27** are configured in the center of the frame. When the anti-rebound device **18** is in the folded position, the car seat attachment points **27** are concealed by the anti-rebound device **18**. Car seat attachment points **27** are configured for attachment of corresponding portions of a car seat thereto, in any manner known to those of skill in the art. Concealing the attachments points **27** before completing the installation of the ISOFIX and anti-rebound device provides additional safety, as the child car seat cannot be installed if the base **10** is not connected or if the anti-rebound device **18** is not extended.

(47) As indicated particularly in FIGS. **1** and **3**, car seat base **10** further includes a series of sensors **11**, **13**, **15**, **17**, **19**, **25**. Each of the sensors is associated with a different functional portion of the car seat base **10**. Sensor **11** is located on or near abutment **16** and determines whether Isofix connectors **34** are latched onto corresponding fittings on the vehicle seat. Sensor **11** may be any type of sensor known to those of skill in the art, for example, a proximity sensor, a magnetic field sensor, a microswitch, or a pressure sensor. In preferred embodiments, sensor **11** is a discrete Hall Effect sensor or a reed switch. As used in the present disclosure, the term “discrete” Hall Effect sensor may also be referred to as a “digital” Hall Effect sensor. A digital Hall Effect sensor outputs two states—off and on. For example, when there is no magnetic field around a flowing current, the state is “off,” and when there is a magnetic field that diverts the current according to the Hall Effect, the state is “on.” A reed switch also operates based on a magnetic field. A reed switch includes a pair of ferromagnetic flexible metal contacts in a hermetically sealed glass envelope. The contacts may be normally open and close in the presence of a magnetic field, or vice versa.

(48) In the operation of sensor **11**, the Isofix connectors may include magnets, such that a magnetic field may be generated when portions of Isofix connectors **34** are displaced to permit entry of the Isofix fittings. Also, in a preferred embodiment, there are two such sensors **11**, one for each Isofix connector **34**.

(49) Sensor **19** is located on or near seat belt guide **36** and determines whether a seat belt has been passed through the seat belt guide and is buckled. Sensor **19** may, for example, be a pressure sensor configured to measure the downward pressure exerted by the seat belt on the seat belt guide **36**. In addition or alternatively, sensor **19** may also be used to determine whether the car seat base **10** is attached to the vehicle seat, even via the seat belt. In preferred embodiments, sensor **19** is a discrete Hall Effect sensor or a reed switch. Anti-rebound device extension sensor **13** determines whether the anti-rebound device **18** is released from its folded position to its extended position. For example, sensor **13** may be an angle or tilt sensor. Alternatively, sensor **13** may monitor the status of a release mechanism of the anti-rebound device **18**. In preferred embodiments, sensor **13** is a discrete Hall Effect sensor or a reed switch. Leg angle sensor **15** senses the angle of telescopic leg **22**, and in particularly whether it is fully extended into the extended position. Leg pressure sensor **19** senses the pressure exerted on the telescopic leg **22** by the floor of the vehicle, to thereby determine whether the telescopic leg has been completely extended. The leg angle sensor **15** and/or leg pressure sensor **19** may also include an analog Hall Effect sensor, which detects an extent of leg extension and folding, based on a measure of voltage generated due to the Hall Effect.
(50) Finally, sensor **25** is configured near car seat attachment points **36**. Sensor **25** may sense whether a car seat is attached to the car seat base **10**. The detection of whether a car seat is attached may be performed with two discrete Hall Effect sensors or reed switches, for example on a left side and right side of the car seat base **10**. Sensor **25** may also include various sensors regarding the position of the car seat and a child restrained therein, such as the orientation of the car seat (e.g., front facing or rear facing), whether the buckles in the car seat are buckled, how the buckles in the car seat are positioned, whether a child is in the car seat, the reclining angle of the car seat, and whether the car seat was subjected to substantial forces and is not suitable for usage or installation. The buckle status sensors and accident sensors may be discrete Hall Effect sensors or reed switches. Certain of the above detection functions, such as installation direction and seat type, may be performed with an RFID reader or an NFC reader, as will be discussed further below.
(51) For example, sensor **25** may include a magnetic field sensor for sensing whether a seat belt of the car seat is buckled. The sensor may include a fixed magnet within the car seat base **10**, and a movable magnet that is attached to a pivotable lever within the frame of the car seat. In an alternative configuration, the pivotable lever and movable magnet are in the frame of car seat base **10**. Buckling the car seat buckle may exert pressure on the pivotable lever (for example, via a spring mechanism), causing the movable magnet to pivot and come into contact with the sensor. Regardless of the location of the movable magnet, a reed switch or discrete Hall Effect sensor is configured within car seat base **10** for sensing the effect of the induced magnetic field.
(52) In another example, sensor **25** may include a magnetic field sensor for sensing whether a child is within the car seat. This magnetic field sensor may also include a fixed magnet within the car seat base **10**, and a movable magnet that is attached to a pivotable lever within the frame of the car seat. The pivotable lever may be locked from pivoting prior to attachment of the car seat to the car seat base **10**. In an alternative configuration, the pivotable lever and movable magnet are configured within car seat base **10**. In this configuration, the pivotable lever may be locked from pivoting prior to installation of the car seat base **10** in a vehicle, for example, by a mechanical bar attached to arms **16**. Installation of the car seat base **10** in the vehicle causes rearward movement of arms **16**, which moves the mechanical bar, releasing the pivotable lever. Placement of a child in the car seat may exert pressure on the lever (for example, via a spring mechanism placed substantially below the center of weight of the child within the car seat), causing the movable magnet to pivot and come into contact with the sensor. A reed switch or discrete Hall Effect sensor is configured within the car seat base **10** for sensing the effect of the induced magnetic field.
(53) One advantage of the embodiments described above is that the buckling status of the car seat and the presence of the child within the car seat may be detected even though the car seat itself does not have any electronics or any power source.

(54) Other sensors, for example, sensors for determining whether the car seat was subjected to substantial forces and is not suitable for usage or installation, may be substantially similar to those described in the above-referenced copending patent applications.
(55) The depicted locations for the sensors are merely schematic, and the sensors may be located at any suitable location for achieving the functions described herein, as can be recognized by those of skill in the art.
(56) The sensors **11**, **13**, **15**, **17**, **19**, **25** may detect both whether a functional portion of the car seat base **10** has been initially installed (e.g., whether anti-rebound device **18** has been fully extended) and also whether an error status has arisen in connection with the monitored functional portion (e.g., whether anti-rebound device **18** is no longer fully extended).
(57) Car seat base **10** further includes a plurality of light emitting elements **26**, **28**, **30**, **32**, **40**. The light emitting elements may be comprised of any material suitable for displaying a light signal, for example, an LED. The LEDs may be of any suitable color and of any suitable dimension. Light emitting element **26** is located on or near abutment **16**. Light emitting element **28** is located on the upper portion **14**, adjacent to push button release mechanism **20**. Light emitting element **30** is located in the proximal portion of the car seat base **10**, near the point of connection of telescopic load leg **22** to frame **12**. Light emitting element **32** is located in the distal section of car seat base **10**, near car seat attachment points **36**. Finally, central light interface **40** is centrally located on the upper portion **14** of the car seat base **10**, in the proximal section, which is an area of high visibility. The depicted locations for the light emitting elements are merely illustrative, and the light emitting elements may be located at any suitable location for achieving the functions described herein, as can be recognized by those of skill in the art.
(58) Electrical console **38** is also configured within frame **12**. Electrical console **38** includes an on-board computer, also referred to herein as a control unit. The on-board computer includes a central processing unit (CPU) with programmable circuitry, also referred to herein as an on-board processor. The depicted location for the on-board computer is merely schematic, and the electrical console may be located at any suitable location within the frame **12**.
(59) The electrical console **38** is electrically connected to the sensors **11**, **13**, **15**, **17**, **19**, **25**, and the light emitting elements **26**, **28**, **30**, **32**, **40**. Through these electrical connections, the on-board computer is configured to receive inputs from the sensors **11**, **13**, **15**, **17**, **19**, **25**, to analyze the inputs, and to instruct the light emitting elements **26**, **28**, **30**, **32**, **40** to emit light signals based on the received inputs, as will be described in further depth below.
(60) Electrical console **38** may also include a communication module and a speaker. The communication module may wirelessly communicate information regarding a status of the sensors, or a status of installation of the car seat base **10**, to a mobile device. The wireless communication can be through any protocol known to those of skill in the art, e.g., Bluetooth or Wi-Fi communication or cellular or any other combination thereof. The speaker may be used to deliver an audible message regarding a status of the sensors or a status of installation of the car seat base. Electrical console **38** may also include a power source, e.g., one or more batteries.
(61) The electrical console **38** may also include a near field communication (NFC) or radiofrequency identification (RFID) reader. The car seat may include an NFC or RFID chip which is readable by the NFC or RFID reader. Advantageously, these technologies enable the car seat to transfer information to the car seat base **10**, without having a power source in the car seat. Exemplary information that may be conveyed via the NFC or RFID chips includes: the type of seat that is connected, specific identification of the seat that is connected (for example, a serial number), direction of the seat (for example, forward or backward facing).
(62) In an alternative embodiment, car seat base **10** is part of a single child restraint system, in which the car seat base **10** is formed as one integral device with the car seat itself. In such embodiments, lights and sensors described above as located on the car seat base **10** may be installed on the car seat instead.

(63) Referring now to FIGS. **5**A-**5**E and **6**A-**6**E, the process by which car seat base **10** guides a user through the installation process will now be described. The car seat base **10** has a specific order, or flow, for the installation steps. Following the designated order ensures that the installation is performed in the simplest way and a shortest time. During the installation process, the sensors **11**, **13**, **15**, **17**, **19**, **25**, light emitting elements **26**, **28**, **30** **32**, **40**, and the processor cooperate to provide guidance to the user as to which installation step to perform next. Optionally, simultaneously, the communication module communicates the progress of the installation to a user's mobile device. The mobile device may run an application that provides a visual instruction—for example, a short video—on how to perform the subsequent installation step.
(64) In the depicted illustration, the installation proceeds in five steps, following bringing of the car seat base **10** to the vehicle seat: (1) attachment of the car seat base **10** to the vehicle, whether via the seat belt or via attachment to Isofix fittings **52**; (2) release and extension of the anti-rebound device **18**; (3) extension of the telescopic load leg **22**; (4) attachment of the car seat to the car seat base **10**; and (5) engaging the child seat buckle when the child is in the seat.
(65) Referring to FIG. **5**A, prior to commencement of the five installation steps, a user begins the installation process by placing the car seat base **10** on a vehicle seat **50**. The base **10** is placed on the seat **52** while telescopic load leg **22** is pivoted into its extended position. As discussed above, the user may grasp the handle **24** or grips while placing the car seat base **10** onto the vehicle seat **52**, so as to cause the load leg **22** to pivot automatically into the extended position, e.g., by the force of gravity.
(66) Sensors detect the progress of the user in placing the car seat base onto the vehicle seat **50**. Such sensors may include, but are not limited to, leg angle sensor **15**, which detects that the telescopic load leg **22** is in the extended position. The sensors provide output to the processor indicating the status of the sensed position car seat base **10**. The processor, upon receiving the output, is adapted to execute code instructions for identifying completion of a current installation step, in this case, placement of the car seat base **10** in position on the vehicle seat **52**.
(67) The processor is further adapted to execute code to instruct light emitting element **26** to emit a light signal. In one exemplary embodiment, the light signal is a solid red LED light. The light signal is emitted at light emitting element **26**, which is on or adjacent to abutment **16**. The light signal is emitted at this location in order to direct the user's attention to the location corresponding to the next installation step, which is attachment of the car seat base **10** to the vehicle seat **52**. Advantageously, the light signal from light emitting element **26** will not light up unless the telescopic load leg **22** is in the unfolded position. Thus, the user will not mistakenly attach the car seat base **10** to the vehicle while the load leg **22** is folded underneath or within the frame.
(68) As seen in FIGS. **5**A-**5**E, central light interface **40** is comprised of light emitting elements arranged in a pattern. In the embodiment of FIGS. **5**A-**5**E, the pattern is a ring of light. The ring of light **40** is divided five sections of light emitting elements, each of which corresponds to an installation step of the car seat base **10**. Section **42** corresponds to attachment of the car seat base **10** to the vehicle **50**; section **44** corresponds to release and extension of the anti-rebound device **18**; section **46** corresponds to extension of the telescopic load leg **22**; section **48** corresponds to attachment of car seat **56** to car seat base **10**; and section **50** corresponds to the buckling of the child in the car seat **56**.
(69) Ring of light **40** is centrally located on the upper section **14** of the car seat base **10**. Unlike light emitting elements **26**, **28**, **30**, **32** which emit light at a location where the user is supposed to perform the next installation step, ring of light **40** is designed to communicate generally the progress of the entire installation process. When a user is supposed to perform a given installation step, i.e., whenever one of the light emitting elements **26**, **28**, **30**, **32** displays a light signal indicating its location, a corresponding section of the ring of light **40** will display a first type of light signal. The light signal may be a solid light in a first color, for example, orange. When the user completes that given installation step, and a sensor senses that the installation step is complete,

the corresponding section of the ring of light **40** will display a second light signal. The second light signal may be a solid light in a second color, for example, green.

(70) The signals delivered by light emitting elements **26**, **28**, **30**, **32** and ring of light **40** may be replicated by other communication techniques. For example, the speaker may deliver audible instructions to the user to perform a particular installation step. Alternatively, the communication module may wirelessly deliver a notification to an application on a user's mobile device. The application may be configured to display the status of the installation of the car seat, for example, by replicating the lights emitted by light emitting elements **26**, **28**, **30**, **32** and ring of light **40**. The application may also show a graphic and/or video instructing the user how to perform each installation step. For example, the graphic and/or video shows attachment of the Isofix connectors **34** to fittings **54** of the vehicle seat **52**.

(71) Referring again to FIG. **5**A, the user then performs the first installation step of attaching the car seat base **10** to the vehicle seat **52**. As discussed above, this step may be performed in two alternative ways: (1) by passing the vehicle seat belt through seat belt guide **36** (not shown in FIG. **5**A) or (2) by attaching Isofix connectors **34** to corresponding fittings **54** of the vehicle seat **52**. Following attachment of the car seat base **10** to the vehicle seat **52**, the sensors detect the attachment. If the car seat is attached with the seat belt, sensor **19** (shown in FIG. **3**) senses the pressure of the vehicle belt on the seat belt guide **36**. If the car seat is attached with the Isofix connectors **34**, sensor **11**, if it is a tension sensor, senses the tension on the connectors **34** or on internal components of abutments **16**. The processor receives outputs from sensors **11** and/or **19**, determines that the attachment step is complete, sends an instruction to light emitting element **26** to stop emitting a signal, and sends an instruction to light emitting element **28** (shown in FIGS. **2-4**) to emit a light signal. Light emitting element **28** is located adjacent to push button release mechanism **20** of anti-rebound device **18**, and thus signals to the user to push the push button release mechanism **20** in order to release the anti-rebound device **18**.

(72) Referring now to FIG. **5**B, correspondingly, light **44** of the ring of light **40**, corresponding to the anti-rebound device **18**, emits the first type of light signal, and light **42** of the ring of light **40** emits the second type of light signal. The user then depresses the push button release mechanism **20** to release the anti-rebound device **18**. Following release and extension of the anti-rebound device **18**, sensor **13** (shown in FIG. **4**) senses the extension of the anti-rebound device **18**. The processor receives an output from sensor **13**, determines that the extension of the anti-rebound device **18** is complete, sends an instruction to light emitting element **28** to stop emitting a signal, and sends an instruction to light emitting element **30** (shown in FIGS. **2-4**) to emit a light signal. Light emitting element **30** is located adjacent to telescopic load leg **22**, and thus signals to the user to extend load leg **22** to the floor of the vehicle.

(73) Referring now to FIG. **5**C, correspondingly, light **46** of the ring of light **40**, corresponding to telescopic load leg **22**, emits the first type of light signal, and light **44** of the ring of light **40** emits the second type of light signal. The user then extends extension section **23** of leg **22** until it is securely pressed against the floor of the vehicle. Following extension of leg **22**, sensor **17** (shown in FIG. **4**) senses the pressure on extension section **23**. The processor receives an output from sensor **17**, determines that the extension of the leg **22** is complete, sends an instruction to light emitting element **30** to stop emitting a light signal, and sends a signal to light emitting element **32** (shown in FIGS. **2-4**) to start emitting a light signal. Light emitting element is located adjacent to car seat attachment points **27** (shown in FIG. **3**), and thus signals to the user to connect car seat **56** to car seat attachment points **27**.

(74) Referring now to FIG. **5**D, correspondingly, light **48** of the ring of light **40**, corresponding to car seat **56**, emits the first type of light signal, and light **46** of the ring of light **40** emits the second type of light signal. The user then attaches the car seat **56** to the car seat base **10**. When sensor **25** (shown in FIG. **4**) detects that the car seat **56** has been correctly attached, it sends an output to the processor. Upon determination by the processor that the car seat has been correctly installed, the

processor sends an instruction to light emitting element **32** to stop emitting a light signal, and to light **48** of the ring of light **40** to emit the second type of light signal.

(75) Referring to FIG. **5**E, correspondingly, light **50** of the ring of light **40**, corresponding to buckling of the car seat **56**, emits the first type of light signal, and light **48** of the ring of light **40** emits the second type of light signal. The user then buckles the child into the car seat **56**. Upon detection by a sensor that the child has been buckled in the car seat, the processor sends an instruction to ring of light **50** to emit the second light signal.

(76) In addition to the above-described functionality during installation of the car seat base **10**, the sensors, light emitting elements, speaker, and communication module may also be used to detect and communicate an error status of the car seat base **10**. For example, in the case of an unsafe situation, the seat may set off a sound alarm and a light alarm. The light alarm may be a blinking light at the closest light emitting element **26**, **28**, **30**, **32**, a blinking light on the central light interface **40**, or a color change on the corresponding portion **42**, **44**, **46**, **48**, **50** of the central light interface **40** (e.g., from green to red). The purpose of the alarms is to instruct the user to rectify the error. Unsafe situations that may provoke an alarm include: (1) installation step not completed correctly or completed in the wrong sequence; (2) child not buckled; (3) one of the parts of the car seat base **10** or car seat **54** not properly connected; (4) car seat **56** no longer structurally safe due to accident impact; and (5) a child left in the seat unattended.

(77) FIGS. **6**A-**6**E depict an alternative embodiment **110** of a car seat base. Unless otherwise specified, car seat base **110** is identical to car seat base **10**, and thus similar elements are designated with similar reference numbers, except that the reference numbers begin with “1.” Car seat base **40** has a central light interface **140**, which depicts a linear pattern, rather than a circular pattern. The user can track the status of the installation by following the successive linear activation of lights in sections **142**, **144**, **146**, **148**, and **150**, in the manner described above in connection with light interface **40**.

(78) FIGS. **7**A-**7**C depict three alternative embodiments for central light interfaces **240**, **340**, and **440**. The central light interfaces **240**, **340**, **440** differ from interfaces **40**, **140** in that they include different patterns for the central light interface. For example, in interface **240**, the fifth section is a linear light interface that runs parallel to the linear interface of the first four sections. Thus, the user can receive assurance that the installation of the car seat base **10** is complete, by completion of the first four sections of the pattern. Then, when the user buckles the child into the car seat, the user can receive separate assurance that the child was properly buckled, by reference to the separate section of the pattern.

(79) Referring now to FIGS. **8**A-**8**B, the mechanics of the extension of anti-rebound device **18** are now discussed. Anti-rebound device **18** includes piston **70**, which may be slidable within sheath **72**. Torsion spring **74** is configured at a point of connection between the piston **70** and the distal end of lower outer section **12** of the frame of car seat **10**. Additionally or alternatively, a helical spring or another plastically deformable device may be concealed within sheath **72** to control the sliding of piston **70** within sheath **72**.

(80) The torsion spring **74** is biased to cause the anti-rebound device **18** to extend backward toward the seat back of vehicle seat **52**. When the anti-rebound device **18** is in the folded position, the push button release mechanism **20** retains the anti-rebound device **18** in place, against the rotational force of the torsion spring **74**. When the push button release mechanism **20** is released, the torsion spring **74** extends automatically backwards, to extend the anti-rebound device **18** through a wide angle and secure the anti-rebound device **18** against the seat back of vehicle seat **52**. When anti-rebound device **18** is fully extended, seat **52** exerts a forward pressure against it, in the direction of arrow **76**.

(81) When the anti-rebound device **18** is released, spring **74** applies force as a normal spring, without any interference from other mechanisms in the car seat base (e.g., push button release mechanism **20**), such that the other mechanisms do not prevent the anti-rebound device **18** from

opening.

(82) Either the spring **74**, or the release mechanism **20**, or both, may further include mechanisms that prevent the anti-rebound device from closing to the folded position, once extended. Such mechanisms may include, but are not limited to, a ratchet or a uni-directional gas piston. In addition, the release mechanism **20** may prevent the anti-rebound device **18** from closing during a rear impact or in response to rebound generated by a frontal impact.

(83) The piston **70** and torsion spring **74** are configured to adaptively adjust their configuration so as to secure the car seat base **10** from rebounding in response to an impact. In particular, the torsion spring **74** may adaptively adjust its tension in response to external forces applied to the vehicle seat back during an impact. In an accident, the car seat may exert force on the vehicle seat **52**, compressing the foam of the vehicle seat **52**, and thereby increasing the space between the anti-rebounding device **18** and the vehicle seat back. Without the adaptive adjustment of torsion spring **74**, the car seat may be jolted backward into the space formerly filled by the seat foam. The adaptive adjustment, however, merely causes the anti-rebounding device **18** to open more, and reduces the bounce back of the car seat **10**. This adaptive adjustment thus fixes the child car seat in place during an accident, and provides an advantage over fixed anti-rebounding devices (such as bars), which lack this capability for adaptive adjustment.

(84) It is possible to design the piston **70** so that it compresses under specific forces and further absorbs energy in a rear impact or rebound effect. When the anti-rebound device is made with such a piston **70**, the piston **70** dampens the forces applied to the child car seat during an impact. Such a mechanism for piston **70** can be realized by several means, including the attachment of the piston **70** to plastically deformable elements that elongate under force.

(85) Another advantage of the adaptive adjustment is realized in the event that the car seat base **10** moves by a small amount. For example, if car seat base **10** is attached to the vehicle seat with a seat belt, and the seat belt tension changes, the car seat base **10** may move slightly relative to the vehicle seat back. In such an instance, the automatic rebound device **18** will adaptively adjust itself to ensure that it remains secured against the vehicle seat back.

(86) Also, during a frontal accident, when the forces applied on the car seat cause the car seat base **10** to slightly move forward, the anti-rebound device **18** will self-adjust and lock the base **10** in place to prevent movement when the car seat rebounds.

(87) FIGS. **9**A-**12**B depict embodiments of a car seat base **510** which include a locking mechanism for anti-rebound device **518**. While the locking mechanism is depicted here in connection with car seat base **510**, it is to be understood that the locking mechanism is compatible in all other respects with features disclosed in connection with car seat base **10**, including the system for advising users as to a progress of installation, the handle or grips, the load leg, and the push button release mechanism.

(88) Referring to FIGS. **9**A-**9**B, anti-rebound device **518** includes a sleeve **571**. FIG. **9**C depicts the locking mechanism with the sleeve **571** removed. Sleeve **571** encloses the locking mechanism, piston **570** and sheath **572**, as shown in FIG. **9**C. Piston **570** and sheath **572** may operate substantially in the same manner as piston **70** and sheath **72** described above, for extending the anti-rebound device between the folded position and the extended position.

(89) As shown in FIG. **9**B, sleeve **571** may be comprised of upper half sheath **571***a* and lower half sheath **571***b*. Upper half sheath **571***a* is able to telescopically slide over the lower half sheath **571***b*, for easy storage when the anti-rebound device **518** is closed.

(90) The locking mechanism includes upper lock arm **576** and lower lock arm **574**. The lower lock arm **574** is slidably received within the upper lock arm **576**. Piston **570** and sheath **572** are substantially parallel to, and mechanically connected to, upper lock arm **576** and lower lock arm **574**. Extension of the piston **570**, e.g., by operation of a spring, correspondingly causes the lower lock arm **574** to extend relative to the upper lock arm **576**. Release pedal **573** is configured within the anti-rebound device **518** above upper attachment point **575**. Extension of piston **570** may also

exert a force on release pedal **573**, to cause it to assume the position shown in FIG. **9**C, in which the locking mechanism is locked.

(91) Sleeve **571** and the elements contained therein (e.g., upper lock arm **576** and sheath **572**) are attached to an upper end of anti-rebound device **518** at upper attachment point **575**. Sleeve **571** is attached to the frame of the car seat base **510** at attachment point **579**, which defines an axis of rotation for the locking mechanism, and the anti-rebound device **518** itself is separately attached to frame of the car seat base **10** at attachment point **577**, which defines an axis of rotation for the anti-rebound device.

(92) Referring now to FIGS. **10**A and **10**B, the upper lock arm **576** includes a tapered receptacle **586**. Tapered receptacle **586** is comprised of two angled surfaces: one parallel to the orientation of lower lock arm **574**, and a second directed at a sharp angle relative to the first surface. The receptacle **586** tapers upwards in a direction from attachment point **579** to release pedal **573**. As used in the present disclosure, the term “narrow end” refers to the end of receptacle **586** that is closer to release pedal **573**, and the term “wide end” refers to the end of receptacle **586** that is closer to attachment point **579**. Angled surface **588** (shown in FIGS. **11**A and **11**B) is further from the lower lock arm **574** at the wide end, and is closer to the lower lock arm **574** at the narrow end.

(93) A roller **583** is held within the receptacle **586** by a spring **585**. Spring **585** forces roller **583** into a position in which it touches at least one angled surface of receptacle **586**, as well as lower lock arm **574**.

(94) In addition, as seen best in FIG. **10**A, release pedal **573** is mechanically connected, through arm **578**, to lever **581**. Lever **581** extends from arm **578** to the narrow end of the receptacle **586**, opposite roller **583**. In the illustrated embodiments, arm **578** is a separate element from sheath **572**. In alternative embodiments, sheath **572** also functions as arm **578**.

(95) FIGS. **11**A and **11**B show a close-up view of the circled area of FIG. **10**B, and in particular demonstrate the operation of the locking mechanism. As shown in FIG. **11**A, when the anti-rebound device **518** is opened, the locking mechanism is also correspondingly opened, and lower lock arm **574** slides through receptacle **586** in the direction of the arrow. Because the roller **583** is in contact with the lower lock arm **574**, this lateral opening of the lower lock arm **574** is translated into a counterclockwise rotational movement of the roller **583**. The roller **583** is consequently pushed in a downward direction toward the wide end of the receptacle **586**. The downward movement of the roller **583** is countered by the force of spring **585**. However, because the wide end of receptacle **586** is wider than the narrow end, there is space for roller **583** to rotate, and thus the rotation of roller **583** does not impede the extension of lock arm **574**.

(96) In FIG. **11**B, the lower lock arm **574** is already extended, and is being pushed upwards relative to upper lock arm **576**. The lateral movement of the lower lock arm **574** is translated to a rotational movement of roller **583** in a clockwise direction. This clockwise movement pushes the roller **583** upward, toward the narrow end of receptacle **586**. Due to the tapering of receptacle **586** and in particular of the angled surface **588**, roller **583** pins lower lock arm **574** in place, and prevents further retraction of the lower lock arm. Indeed, further retraction of the lower lock arm **574** only increases the strength of the locking action, because it causes roller **583** to be inserted further into the narrow end. The upward force of spring **585** ensures that constant friction is maintained between roller **583**, angled surface **588**, and lower lock arm **574**.

(97) FIG. **12**A depicts release of the release pedal **573**, and FIG. **12**B is a close up view of the circled area of FIG. **12**A. When release pedal **573** is pulled forward and downward, arm **578** is correspondingly pushed downward, via an eccentric mechanism. Depression of release pedal **573** causes lever **581** to displace roller **583** from its locked position in the narrow end (indicated by the dashed circle in FIG. **12**B) toward the wide end. Once the roller **583** is displaced, lower lock rod **574** is again able to slide relative to upper lock rod **576**, permitting the anti-rebound device **518** to close.

(98) Reference is now made to FIGS. **13**A to **15**C, which depict the mechanics of push button

release mechanism **20**, and the interrelationship of push button release mechanism **20** with Isofix connectors **34** and seat belt guide **36**. In order to make the installation process more intuitive, and to prevent improper installation of car seat base **10**, it is advantageous to prevent release of the anti-rebound device **18** until the car seat base **10** is first properly secured to the vehicle. It is further advantageous to utilize a single mechanism for preventing and for permitting release of the anti-rebound device **18**, regardless of how the car seat base **10** is secured to the vehicle—i.e., whether by the Isofix system or by the seat belt. The mechanism depicted in FIGS. **13**A-**15**C provides this functionality.
(99) FIG. **13**A depicts a state in which the push button release mechanism **20** is in a locked position, and cannot be pushed down. In this configuration, button stopper **80** is configured beneath push button **20**, and mechanically prevents depression of the push button release mechanism **20**. FIG. **13**B depicts a state in which button stopper is mechanically retracted relative to its configuration in FIG. **13**A, thereby allowing depression of push button release mechanism **20**.
(100) Rod **82** is mechanically connected to the button stopper **80**, at a proximal end of the rod **82**. At a distal end of the rod **82**, the rod **82** is mechanically connected to both (1) a horizontal connector **85** between abutments **16** for Isofix connectors **34** and (2) a seat belt pedal **90**. The Isofix connectors **34** are adapted to be attached to corresponding fittings on the vehicle seat, as discussed above. The seat belt pedal **90** is adapted to be located beneath seat belt **92**, which is passed through seat belt guide **36** (not shown in FIGS. **13**A-**13**B) and buckled to a buckle on the vehicle seat, as discussed above.
(101) As depicted in FIGS. **13**A, **13**B, and **15**A, the rod **82**, and correspondingly the button stopper **80** attached thereto, may be retracted through either connection of the Isofix connectors **34** or buckling of the seat belt **92**. With respect to connection of the Isofix connectors **34**, each of the abutments **16** includes a spring **84** and a corresponding spring-loaded piston **86**. A first pivot **88** is configured between each of the abutments **16** and the rod **82**. The pivot **88** is fixedly connected to the rod **82**, but has a pivot point **83** at the intersection of pivot **88** and horizontal connector **85**. When the Isofix connectors are closed, springs **84** push spring-loaded pistons **86**, to cause pistons **86** to move proximally. The proximal movement of pistons **86** causes pivot **88** to pivot in a forward direction, and thereby causes corresponding rearward movement of the rod **82**.
(102) With respect to buckling of the seat belt **92**, belt pedal **90** is attached to rod **82** via a second pivot **96**. Belt pedal **90** includes extension **94**. When belt pedal **90** is depressed by the force of the buckled seat belt **92**, extension **94** is driven into pivot **96**. This causes second pivot **96** to pivot in a forward direction, and causes corresponding rearward movement of the rod.
(103) In an advantageous embodiment, substantially identical rearward movement of the rod **82** is caused by either buckling the seat belt **92** or by attaching the Isofix connectors **34** to the corresponding fittings of the vehicle seat. Advantageously, the push button release mechanism is thus released in an identical way, regardless of the method employed to attach the car seat base to a vehicle seat.
(104) As discussed above in connection with FIGS. **2-4**, in the illustrated embodiment, the anti-rebound device **18** is released by depression of a push button release mechanism **20** following attachment of the car seat base **10** to the vehicle. In alternative embodiments, the anti-rebound device **18** is extended automatically upon attachment of either Isofix connectors **34** or a seat belt to the vehicle. In such embodiments, the rod **82** is itself connected to the release mechanism, rather than to a button stopper that prevents depression of a push button, so that movement of the rod itself releases the release mechanism.
(105) While the foregoing written description of the invention enables one of ordinary skill to make and use what is considered presently to be the best mode thereof, those of ordinary skill will understand and appreciate the existence of variations, combinations, and equivalents of the specific embodiment, method, and examples herein. The invention should therefore not be limited by the above described embodiment, method, and examples, but by all embodiments and methods within

the scope and spirit of the invention as claimed.
(106) It is expected that during the life of a patent maturing from this application many types of sensors, many types of light emitting devices, and many types of push button release mechanisms will be developed and the scope of the terms sensor, light emitting device, and push button release mechanism is intended to include all such new technologies a priori.
(107) As used herein the term “about” refers to ±10%.
(108) The terms “comprises”, “comprising”, “includes”, “including”, “having” and their conjugates mean “including but not limited to”. This term encompasses the terms “consisting of” and “consisting essentially of”.
(109) The phrase “consisting essentially of” means that the composition or method may include additional ingredients and/or steps, but only if the additional ingredients and/or steps do not materially alter the basic and novel characteristics of the claimed composition or method.
(110) As used herein, the singular form “a”, “an” and “the” include plural references unless the context clearly dictates otherwise. For example, the term “a compound” or “at least one compound” may include a plurality of compounds, including mixtures thereof.
(111) The word “exemplary” is used herein to mean “serving as an example, instance or illustration”. Any embodiment described as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments and/or to exclude the incorporation of features from other embodiments.
(112) The word “optionally” is used herein to mean “is provided in some embodiments and not provided in other embodiments”. Any particular embodiment of the invention may include a plurality of “optional” features unless such features conflict.
(113) Throughout this application, various embodiments of this invention may be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the invention. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 3, 4, 5, and 6. This applies regardless of the breadth of the range.
(114) Whenever a numerical range is indicated herein, it is meant to include any cited numeral (fractional or integral) within the indicated range. The phrases “ranging/ranges between” a first indicate number and a second indicate number and “ranging/ranges from” a first indicate number “to” a second indicate number are used herein interchangeably and are meant to include the first and second indicated numbers and all the fractional and integral numerals therebetween.
(115) It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable subcombination or as suitable in any other described embodiment of the invention. Certain features described in the context of various embodiments are not to be considered essential features of those embodiments, unless the embodiment is inoperative without those elements.
(116) All publications, patents and patent applications mentioned in this specification are herein incorporated in their entirety by reference into the specification, to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated herein by reference. In addition, citation or identification of any reference in this application shall not be construed as an admission that such reference is available as prior art to the present invention. To the extent that section headings are used, they should not be construed as

necessarily limiting. In addition, any priority document(s) of this application is/are hereby incorporated herein by reference in its/their entirety.

Claims

1. A system for guiding installation of a car seat base and a corresponding car seat, comprising: a plurality of sensors mounted on the car seat base for detecting a progress of a user in a plurality of sequential installation steps required for installing the car seat base and the corresponding car seat on a vehicle seat; a plurality of light emitting elements arranged at various locations on the car seat base, each location corresponding to one of the plurality of installation steps; a processor adapted to execute code instructions for: in response to outputs a first light signal of at least one of the plurality of sensors, identifying completion of a current installation step of the plurality of installation steps; and instructing a suitable light emitting element from the plurality of light emitting elements to output a second light signal indicate a location corresponding to an installation step following the current installation step, wherein, upon detection of an error status by one of the sensors, one or more of the plurality of light emitting elements is configured to display an error light signal to instruct the user to rectify the error, wherein the error status comprises at least one of: (1) installation step not completed correctly or completed in the wrong sequence; (2) child not buckled; (3) one of the parts of the car seat base or car seat not properly connected; and (4) car seat no longer structurally safe due to accident impact.
2. The system of claim 1, wherein the car seat base further comprises a central light interface comprised of light emitting elements arranged in a pattern, each of the light emitting elements of the central light interface corresponding to one of the installation steps, wherein, for each of the installation steps, whenever a light emitting element from the plurality of light emitting elements indicates a location corresponding to an installation step.
3. The system of claim 1, wherein the car seat base further comprises a communication module configured to wirelessly deliver information to a mobile device about a status of each of the installation steps and/or an alert status, and further comprising code instructions embodied in a non-transitory medium stored on the mobile device, the code instructions configured, upon receipt of a status of completion of each installation step, to display a pictorial representation of a progress of installation of the car seat base and instructions for performing a subsequent step of the sequential installation steps.
4. The system of claim 1, wherein the car seat base further comprises a speaker for delivering audible messages regarding a status of each of the installation steps and/or an alert status.
5. The system of claim 1, wherein the plurality of sensors comprise sensors for detecting one or more of: (1) leg angle of a telescopic load leg; (2) pressure on the telescopic load leg; (3) extension of the telescopic load leg; (4) whether an Isofix connector of the base is engaged with a corresponding child safety seat attachment point of the vehicle; (5) whether a seat belt is passed through a seat belt guide; (6) whether an anti-rebound device is in a folded position or an extended position; (7) orientation of the car seat as front facing or rear facing; (8) whether a car seat belt buckle is engaged; (9) presence of a child in the car seat; (10) reclining angle of the car seat; (11) type of car seat connected to the base; (12) whether the car seat was subjected to substantial forces and is not suitable for usage or installation; and (13) whether a power source for the car seat base has sufficient power.
6. The system of claim 1, wherein the car seat base further comprises a frame, a load leg that is pivotable from a folded position beneath the frame to an extended position between the frame and a vehicle floor, and at least one handle or grip oriented on the frame to enable grasping of the frame, wherein, when the car seat base is grasped from the at least one handle or grip, the leg pivots into the extended position.
7. The system of claim 6, wherein the at least one handle or grip is configured at least one of (1) on

an upper portion of the frame; (2) on a front portion of the frame; (3) in a rear portion of the frame; and (4) on two parallel side portions of the frame.

8. The system of claim 6, further comprising a locking mechanism for locking the leg into the extended position.

9. The system of claim 1, wherein the car seat base has a release mechanism rod mechanically connected to (1) a stopper preventing release of the release mechanism; (2) a belt pedal adapted to be located beneath a seat belt; and (3) two arms each having a connecting latch belt.